LIONSCRAFT

A TRADFI PRIMER

DeFi Vaults

Programmable funds, explained.

THE BIG IDEA

What is a DeFi vault?

A smart contract that pools user deposits and runs an automated investment strategy on their behalf.

Think of it as a **programmable mutual fund** — no custodian, no transfer agent, no fund administrator. The code is all three.

RHYMES WITH

Mutual fund / ETF	Pooled capital, professional strategy
Money market fund	Yield on idle cash, daily liquidity
Managed account	Strategy runs; you hold the shares
Closed-end fund	When redemptions are gated

MECHANICS

How it works, end to end

STEP 01

1

Deposit

User sends tokens (e.g. USDC) to the vault contract.

STEP 02

2

Mint

Vault issues tokenized shares representing pro-rata ownership.

STEP 03

3

Deploy

Code puts capital to work: lending, LPing, staking, etc.

STEP 04

4

Earn

Share value rises as the strategy generates returns.

STEP 05

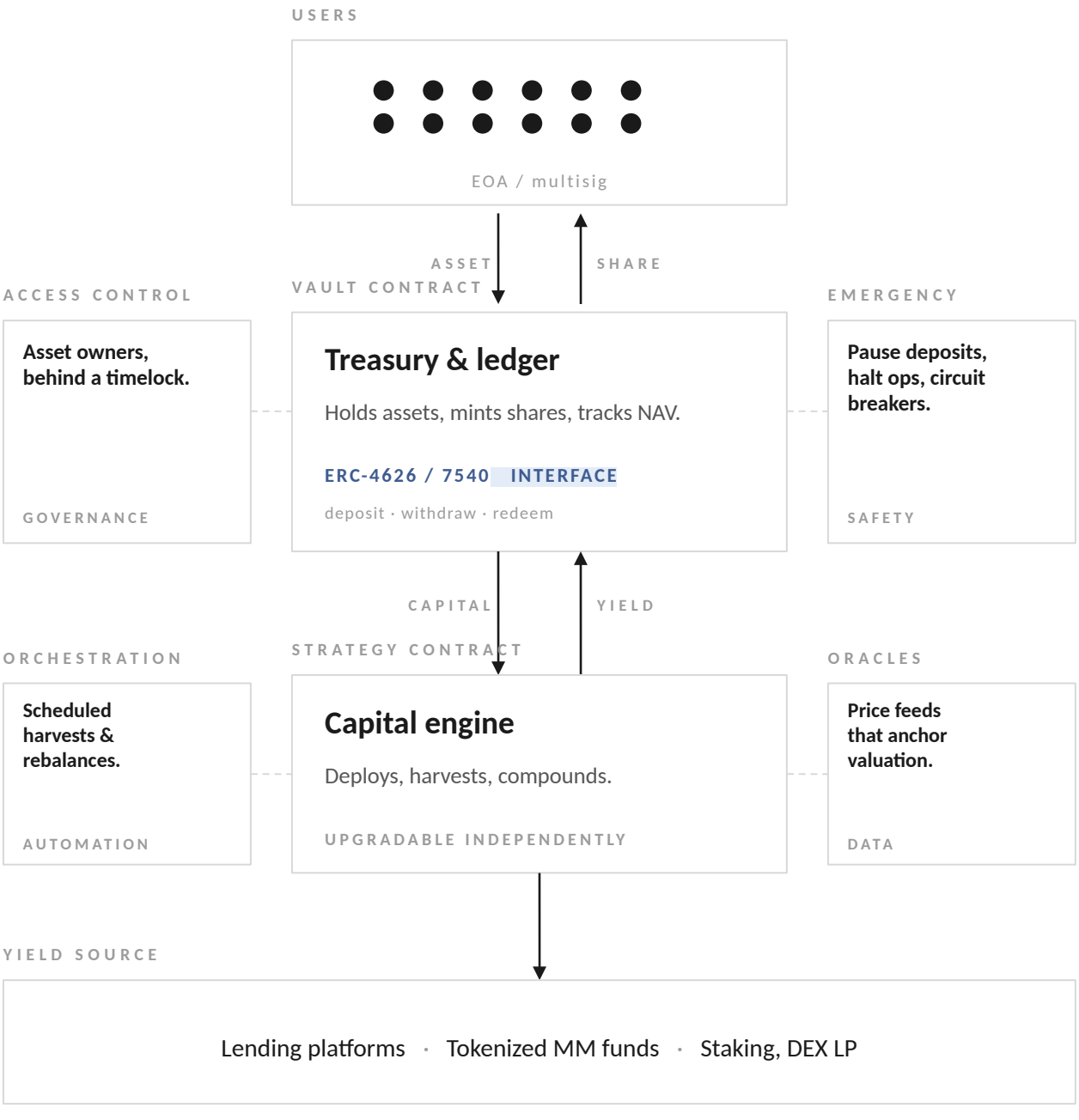
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Redeem

Burn shares → receive underlying + earned yield.

EVERYTHING ON-CHAIN · deposits, share issuance, strategy execution, and redemptions are all smart-contract calls.

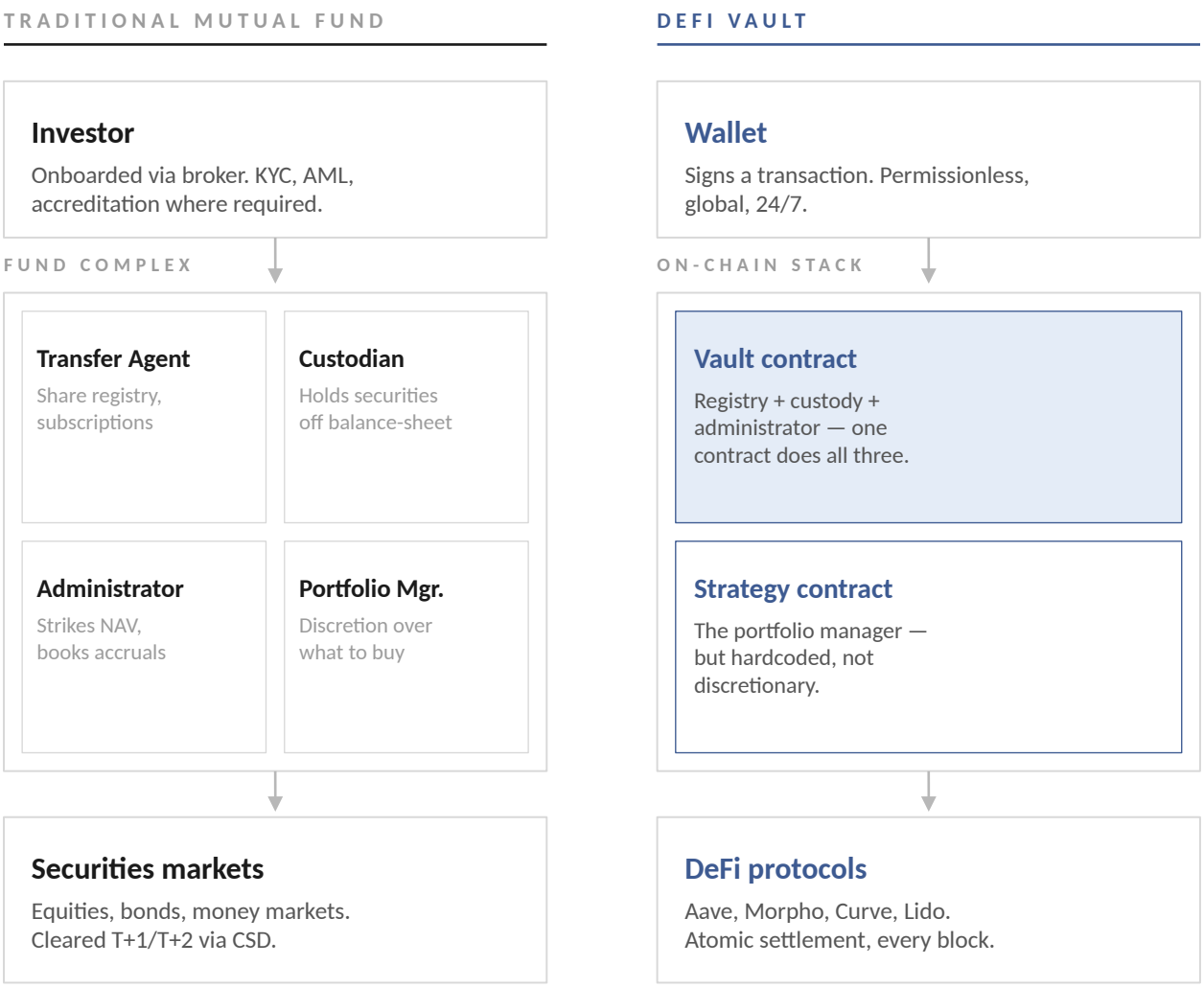
Architecture at a glance



SAME PLUMBING, DIFFERENT STACK

The fund complex, rebuilt as code

Every TradFi middle-office role has a DeFi counterpart. The box just moves from an org chart to a contract.



BOTTOM LINE · Four separate service providers collapse into two contracts. **Fewer handoffs, more code risk.**

EXAMPLE IN ACTION

$10,000 for a year, two ways

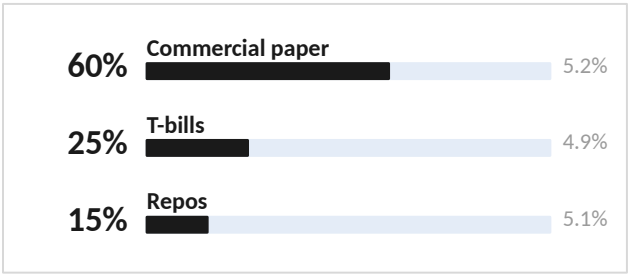
Same capital, same risk bucket, same horizon — follow the cash through a mutual fund and a DeFi vault.

MONEY MARKET FUND

YOU SUBSCRIBE

$10,000 → **10,000 shares**
@ $1.000 NAV · 4pm cutoff · T+1

FUND HOLDINGS



FUND AT T+1 YEAR

NAV: **$1.044** (was $1.000)
After 30 bps expense ratio

YOU REDEEM

10,000 shares → **$10,440**
Wired next business day

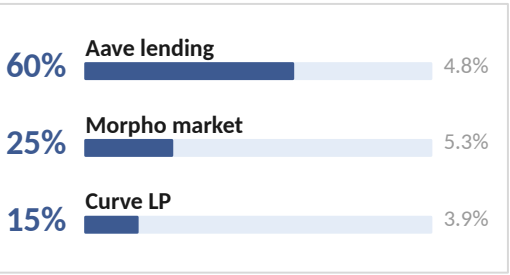
NET YIELD
4.40%

USDC VAULT (yvUSDC)

YOU DEPOSIT

10,000 USDC → **10,000 yvUSDC**
@ $1.000 PPS · atomic, same block

STRATEGY ALLOCATION



VAULT AT T+1 YEAR

PPS: **$1.047** (was $1.000)
After 10% performance fee

YOU REDEEM

10,000 yvUSDC → **10,470 USDC**
Atomic, same block

NET YIELD
4.70%

SETTLEMENT · T+1 wire, 9am–4pm ET cutoff vs. **atomic, 24/7, no minimum**

Vault yields ~30 bps higher in this comparison — but for this gap you're trading audited fund-administrator oversight for smart-contract risk.

COMPONENTS

The four moving parts

01

01

The vault contract

Smart contract that holds assets, mints shares, and enforces the rules. Immutable unless an upgrade path is explicit.

02

02

Tokenized shares

Your claim on the pool, represented as an ERC-20 token. Transferable, collateralizable, composable with the rest of DeFi.

03

03

The strategy

The logic that deploys deposits to earn yield — lending, market-making, staking, arbitrage, tokenized credit.

04

04

A standard interface

ERC-4626 and its successors let wallets, aggregators, and other protocols plug in uniformly — see next slide.

Standards: why vaults plug into everything

Three ERCs define the vault surface. Each one solves a different problem the base standard couldn't.

ERC-4626

Tokenized Vault

The base standard — synchronous single-asset vaults.

- Single underlying asset, single share token.
- deposit / mint / withdraw / redeem — all instant.
- Adopted by Yearn, Morpho, Aave, most yield protocols.

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ERC-7540

Asynchronous Vault

Request-based deposits & redemptions.

- Users request; operator settles in a later block.
- Needed for RWAs, rate-limited strategies, cross-chain.
- Unlocks BlackRock BUIDL, Centrifuge, Ondo-style products.

LIVE · 2024

ERC-7575

Multi-Asset Vault

Share token decoupled from the vault contract.

- One share representing baskets or multi-collateral pools.
- Enables LP-style vaults with multiple underlyings.
- Additive — pairs cleanly with 4626 and 7540.

FINAL · 2025

YIELD SOURCES

Where does the yield come from?

Vaults don't manufacture returns. They route capital to real on-chain activity.

01

Lending interest

Borrowers on Aave, Morpho, Compound pay rates to depositors.

02

Trading fees

LPs on Uniswap, Curve, Balancer earn a cut of every swap.

03

Staking rewards

Validators earn protocol issuance for securing the chain.

04

Incentive tokens

Protocols pay in their own tokens to bootstrap liquidity.

05

Arbitrage / MEV

Strategies capture price gaps between venues and pools.

06

Real-world assets

T-bills and tokenized credit — fastest-growing category.

GEGENÜBERSTELLUNG

Role by role: who does what?

Every service provider in a fund complex maps to something in the vault's code.

TRADFI ROLE	HOW IT WORKS	DEFI EQUIVALENT	HOW IT WORKS
Transfer Agent	Maintains the register of who owns which shares.	**ERC-20 token**	Balances live in the token contract — visible to anyone.
Custodian	Holds the portfolio's securities on client behalf.	**Vault contract**	Assets sit in the contract's address. No outside custodian.
Administrator	Strikes NAV, books accruals, runs the fund's books.	**Vault accounting**	totalAssets() / pricePerShare() computed every block.
Portfolio Manager	Discretionary decisions on what to buy and sell.	**Strategy contract**	Logic is hardcoded or governed — no discretion at runtime.
Prime Broker	Provides leverage, securities lending, clearing.	**Lending protocol**	Aave / Morpho markets — borrow, post collateral, liquidate.
Market Maker	Quotes prices, provides liquidity in fund shares.	**DEX / AMM**	Constant-function pools; anyone can trade or provide liquidity.
Pricing Service	Provides marks for illiquid or OTC positions.	**Oracle**	Signed price feeds (Chainlink, Pyth) written on-chain.
Auditor	Periodic post-hoc review of books and controls.	**Code audit + on-chain**	Pre-deploy audits + continuous public verification.

THE RE-COMPOSITION · Four service providers compress into two contracts — and the auditor's job gets continuous instead of periodic.

SIDE BY SIDE

What's different from a mutual fund?

	TRADITIONAL MUTUAL FUND	DEFI VAULT
CUSTODY	Fund administrator holds assets	**Smart contract holds; user holds keys**
ACCESS	Accredited / KYC'd investors	**Any wallet, permissionless, global**
MINIMUMS	$1k–$25k typical; institutional tiers	**No minimum — dust to millions**
MARKET HOURS	9:30–4 ET; cut-off for same-day NAV	**24 / 7 / 365, same behavior globally**
SETTLEMENT	T+1 / T+2 via CSD, wire instructions	**Atomic — same block as the call**
LIQUIDITY	End-of-day NAV, redemption gates	**Per-block (or async via ERC-7540)**
TRANSPARENCY	Quarterly holdings disclosure, lag	**Every position visible on-chain**
DISTRIBUTIONS	Periodic cash dividend, 1099 forms	**Auto-compound in share price**
CROSS-BORDER	FX, wrapper funds, local distributors	**Same wallet, any country, one click**
MANAGER	PM with discretion	**Code — hardcoded or governed**
FEE MODEL	Mgmt + performance, opaque netting	**Stated in the contract, typically lower**

NOT A FREE LUNCH

The risks TradFi folks should know

R 1

Smart contract risk

A bug in the vault or any protocol it touches can drain the pool. Audits reduce — don't eliminate — this.

R 2

Economic design risk

Strategies can unwind in stress (depegs, cascading liquidations, oracle lag). Backtests lie under novel conditions.

R 3

Counterparty / protocol risk

Vaults stack on other protocols. If a dependency fails, the vault inherits the loss.

R 4

Governance & key risk

Upgradeable contracts and multisigs mean humans can still move funds. Read the privilege surface.

R 5

Regulatory ambiguity

Legal status varies by jurisdiction and is evolving. Securities, AML, and tax treatment are all unsettled.

TAKEAWAYS

The one-line summary

01

Vaults are programmable funds.

Pooled capital + automated strategy + tokenized shares — all on-chain.

02

The code is the middle office.

No custodian, administrator, or transfer agent. This is the cost savings and the risk.

03

Yield is real; so are the failure modes.

Returns come from lending, trading, staking, and RWAs — with smart-contract and design risk on top.

04

Standards are turning vaults into infra.

ERC-4626 + 7540 + 7575 make vaults plug-and-play building blocks for the rest of DeFi.

GET IN TOUCH

Let's continue the conversation.



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This deck is a primer, not investment advice. Vault selection depends on your mandate, jurisdiction, and operational setup — happy to dig in.